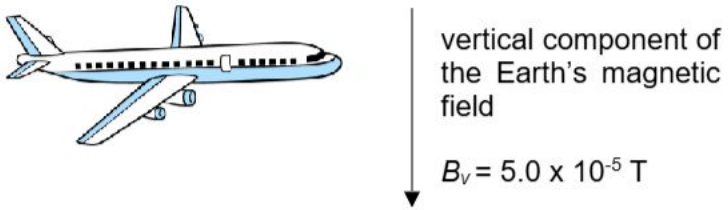
25 A plane heads East at a speed of 30 m s^{-1}, cutting the vertical component of the Earth's magnetic field of flux density B_V = 5.0 x 10^{-5} T acting downwards



Taking the wingspan of the plane to be 17 m, what is the e.m.f generated across the wing and which side of the wing will be at higher potential?

	E.m.f generated/mV	Wing which is at higher potential
A	0	None
B	0.030	Left
C	26	Right
D	26	Left

Ans: D

E.m.f. generated, $E = BIv = \left(5.0 \times 10^{-5}\right)(17.0)(30) = 0.025\ \text{V} = 26\,\text{mV}$

By Lenz's Law, the induced current must be flowing from the right wing to the left wing (FLHR) such that the magnetic effect will oppose the change.

Therefore, the *left* wing is at a higher potential.